SC Honors Geometry - Nappier

5.5 Proof Practice — CPCTC

Every proof here is two jobs. FIRST prove the triangles congruent, exactly as you did on Day 1. THEN — and only then — take the part you were asked about, because corresponding parts of congruent triangles are congruent. CPCTC is never line 1, and it is never the reason the triangles are congruent.

REASON BANK — every reason on this page is one of these

Given · Reflexive Property of Congruence · Definition of midpoint · Definition of a segment bisector · Definition of angle bisector

Vertical Angles Theorem · Alternate Interior Angles Theorem · Definition of perpendicular lines · Right Angles Congruence Theorem · Definition of a right triangle

SSS Congruence Theorem · SAS Congruence Theorem · ASA Congruence Theorem · AAS Congruence Theorem · HL Congruence Theorem · CPCTC

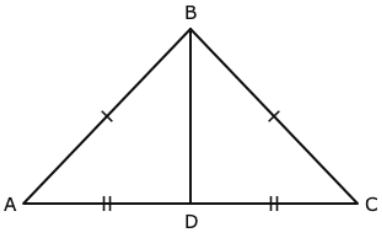
Part A — The statements are done. Supply each reason.

1. A part, once the triangles are yours (5.5)

Given: $A\bar{B} \cong C\bar{B}$, and D is the midpoint of $A\bar{C}$

Prove: $\angle A \cong \angle C$

The last line is an ANGLE. Everything above it is about triangles.



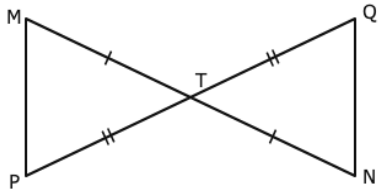
Statements	Reasons
1. $A\bar{B} \cong C\bar{B}$	**1.**
2. D is the midpoint of $A\bar{C}$	**2.**
3. $A\bar{D} \cong C\bar{D}$	**3.**
4. $B\bar{D} \cong B\bar{D}$	**4.**
5. $\triangle ABD \cong \triangle CBD$	**5.**
6. $\angle A \cong \angle C$	**6.**

2. Across the crossing (5.5)

Given: $M\bar{N}$ and $P\bar{Q}$ bisect each other at T

Prove: $M\bar{P} \cong N\bar{Q}$

MP and NQ are not given and not shared. They have to be earned.



Statements	Reasons
1. $M\bar{N}$ and $P\bar{Q}$ bisect each other at T	**1.**
2. $M\bar{T} \cong N\bar{T}$ and $P\bar{T} \cong Q\bar{T}$	**2.**

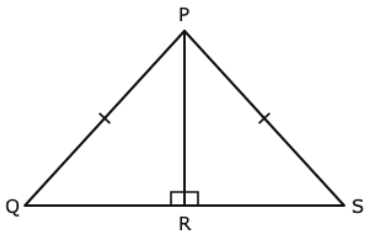
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Statements	Reasons
3. $\angle MTP \cong \angle NTQ$	**3.**
4. $\triangle MTP \cong \triangle NTQ$	**4.**
5. $M\overline{P} \cong N\overline{Q}$	**5.**

3. A right triangle, then the base (5.5)

Given: $P\overline{R} \perp Q\overline{S}$, and $P\overline{Q} \cong P\overline{S}$
Prove: $Q\overline{R} \cong S\overline{R}$

This is the proof that an isosceles triangle's altitude bisects its base.



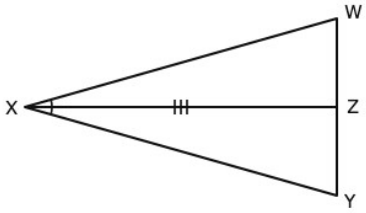
Statements	Reasons
1. $P\overline{R} \perp Q\overline{S}$	**1.**
2. $\angle PRQ$ and $\angle PRS$ are right angles	**2.**
3. $\triangle PQR$ and $\triangle PSR$ are right triangles	**3.**
4. $P\overline{Q} \cong P\overline{S}$	**4.**
5. $P\overline{R} \cong P\overline{R}$	**5.**
6. $\triangle PQR \cong \triangle PSR$	**6.**
7. $Q\overline{R} \cong S\overline{R}$	**7.**

Part B — The reasons are done. Write each statement.

4. An angle you could not see (5.5)

Given: $X\overline{Z}$ bisects $\angle WXY$, and $\angle W \cong \angle Y$
Prove: $W\overline{Z} \cong Y\overline{Z}$

Finish the triangle congruence first. The side comes last, for free.



Statements	Reasons
1.	**1.** Given
2.	**2.** Definition of angle bisector
3.	**3.** Given

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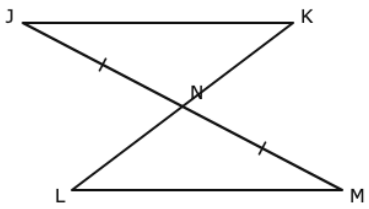
Statements	Reasons
4.	**4.** Reflexive Property of Congruence
5.	**5.** AAS Congruence Theorem
6.	**6.** CPCTC

5. Parallel, proved rather than given (5.5)

Given: $\overline{JK} \parallel \overline{ML}$, and N is the midpoint of $\overline{JM}$

Prove: $\overline{KN} \cong \overline{LN}$

You already proved these triangles congruent on Day 1. Add one line.



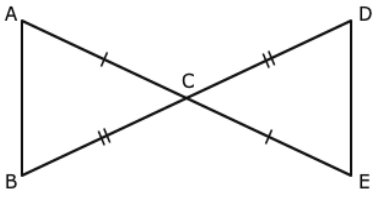
Statements	Reasons
1.	**1.** Given
2.	**2.** Alternate Interior Angles Theorem
3.	**3.** Given
4.	**4.** Definition of midpoint
5.	**5.** Vertical Angles Theorem
6.	**6.** ASA Congruence Theorem
7.	**7.** CPCTC

6. Two sides across a point, one angle out (5.5)

Given: $\overline{AC} \cong \overline{EC}$, and $\overline{BC} \cong \overline{DC}$

Prove: $\angle A \cong \angle E$

Name the triangles in an order that puts A and E in matching places.



Statements	Reasons
1.	**1.** Given
2.	**2.** Given
3.	**3.** Vertical Angles Theorem
4.	**4.** SAS Congruence Theorem

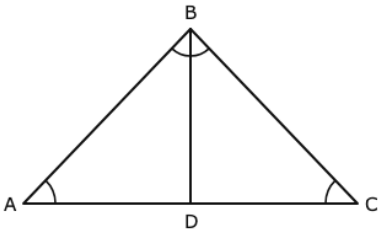
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Statements	Reasons
5.	**5.** CPCTC

Part C — On your own. Statements AND reasons.

7. The isosceles base angles (5.5)

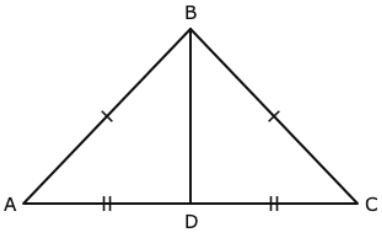
Given: $A\bar{B} \cong C\bar{B}$, and $B\bar{D}$ bisects $\angle ABC$
Prove: $\angle A \cong \angle C$
This is the Base Angles Theorem, proved from scratch.



Statements	Reasons
1.	**1.**
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**

8. All the way to a right angle (5.5)

Given: $A\bar{B} \cong C\bar{B}$, and D is the midpoint of $A\bar{C}$
Prove: $B\bar{D} \perp A\bar{C}$
CPCTC gets you congruent angles. A linear pair of congruent angles is a right angle pair — say so, then name the perpendicular.



Statements	Reasons
1.	**1.**
2.	**2.**
3.	**3.**
4.	**4.**
5.	**5.**
6.	**6.**

5.5 Proof Practice — CPCTC

Statements	Reasons
7.	**7.**
8.	**8.**